

An-Chi Liu

SENIOR SOFTWARE ENGINEER · AI PLATFORMS, RAG & CLOUD-NATIVE INFRASTRUCTURE

Taiwanese, based in Tokyo, Japan (Permanent Resident)

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Summary

Engineer who starts platforms from zero and gets them adopted: built T2's company-wide **evaluation and QA platform** on hybrid on-premises **GPU** and cloud **Kubernetes**, co-founded an **LLM/RAG** product, and co-maintain a 3.1k-star **C++** networking library. Decides by measurement: model, hosting, and vendor choices made on evaluated cost, latency, and quality. Network-native background (Ethernet/IP, packet capture, sub-50 ms streaming); award-winning M.S. thesis on cloud offloading.

Work Experience

T2, Inc. (L4 Autonomous Trucking)

Senior Software Engineer

Tokyo, Japan

Nov. 2025 – Present

- **Conceived, proposed, and delivered** a company-wide **evaluation and QA platform** from zero: wrote the design, secured budget, drove adoption through user discovery. Now runs hundreds of jobs per day for the whole autonomy stack.
- Chose **hybrid Kubernetes** over pure cloud: on-premises **GPU** nodes for steady load, cloud workers for bursts, cutting per-run cost and keeping proprietary data on site. **Go** backend, **React** frontend, automated **Docker** delivery.
- Built a **validation framework** that scores millions of runs against explicit metrics. Replaced thousands of engineer-hours of manual review and forms the evidence base for T2's **safety certification**.
- Integrated the platform with internal services and **CI** so every code change is regression-tested automatically before merge.

Datra AI

Technical Co-founder

Remote

Feb. 2025 – Sep. 2025

- Co-founded an **LLM** product for **Shopify** merchants. Owned architecture end to end (**Go**, **React**, **PostgreSQL**, **AWS**, **Docker**) and managed two developers.
- Designed the **RAG** pipeline: merchant data embedded into **pgvector**, retrieval-grounded prompting, and a labeled test set to measure answer quality and catch hallucinations before release.
- Evaluated **OpenAI** and **Claude** models per task on cost, latency, and quality; routed each workflow to the cheapest model that met its bar and kept providers swappable to avoid lock-in.

Mujin, Inc.

Backend Software Engineer

Tokyo, Japan

Sep. 2022 – May 2025

- Built a live-streaming and playback system for **100+ cameras** with under **50 ms** end-to-end latency using **WebRTC**, **FFmpeg**, and **Go**, selected after benchmarking against RTSP and HLS alternatives.
- Developed a **C++ network discovery service** (PcapPlusPlus) that locates **Ethernet/IP** robots, industrial cameras, and edge devices across factory networks within seconds.
- Extended production **C++/Python** robot and warehouse control systems (real-time motion, sensor, manipulation) and exposed them through Go web and C++ ZMQ APIs.

Education

National Yang Ming Chiao Tung University (NYCU)

M.S. in Computer Science and Engineering

Hsinchu, Taiwan

Sep. 2019 – May. 2022

- GPA: 4.11/4.3. Thesis: **Offworker: An Offloading Framework for Parallel Web Applications**, published in **WISE 2022** (top-tier conference).
- Received the **Best Master's Thesis Award** from the Institute of Information & Computing Machinery, Taiwan (2022; one of 13 awardees).

National Taiwan University (NTU)

B.S. in Bio-Industrial Mechatronics Engineering

Taipei, Taiwan

Sep. 2014 – Jun. 2019

Open-Source Projects

solvcon

- **Major contributor** to a **C++/Python** toolkit for PDE simulation, numerical computing, and AI-assisted engineering: C++ time-series kernels, DataFrame operators, MCAP/ROS 2 ingestion. [PRs]
- Built the toolkit's **AI agent** system: an LLM backend with tool calling that drives the Pilot GUI and numerical workflows. Presented at **PyCon JP 2024**.

PcapPlusPlus

- **One of five core maintainers** since 2024 for a 3.1k-star, multi-platform **C++** packet-processing library.
- Improved performance, modernized the **CI/CD** pipeline, added protocol support, and expanded documentation.

Skills

AI / LLM

LLM application architecture, RAG, pgvector, embeddings, prompt engineering, model evaluation, OpenAI/Anthropic APIs, AI agents

Cloud / Platform

Kubernetes (hybrid on-prem GPU + cloud), Docker, AWS, Linux, CI/CD, GPU job scheduling, evaluation pipelines, PostgreSQL

Networking

Ethernet/IP, packet capture and protocol parsing (PcapPlusPlus maintainer), WebRTC, ZMQ, low-latency streaming

Programming

Proficient: Python 3, C++11/14/17, Go; **Experience:** TypeScript, JavaScript, Bash

Languages

Mandarin Chinese (Native), English (Business), Japanese (Limited Conversational)